LAB SHEET - 09

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SECTION: B

ROLL.NO.: 53

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Data Storytelling and Business Insight Generation using Interactive Visualizations

Aim

To create a compelling data-driven story by analyzing a real-world dataset, developing meaningful visualizations, and presenting actionable business insights to support strategic decision-making.

Dataset

Netflix Movies and TV Shows Dataset

```
%pip install -q pandas numpy matplotlib seaborn plotly kaleido==0.2.1
```

```
                                        79.9/79.9 MB 4.7 MB/s eta 0:00:00
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
import plotly.graph_objects as go

from plotly.subplots import make_subplots
from IPython.display import display, HTML

pd.set_option("display.max_columns", None)

sns.set_theme(style="whitegrid")

print("Libraries imported successfully.")
```

```
Libraries imported successfully.
```

```
from IPython.display import display, Image
import plotly.io as pio

pio.renderers.default = "colab"

print("Libraries imported successfully.")
```

```
Libraries imported successfully.
```

Step 1: Select a Real-World Dataset

The Netflix Movies and TV Shows Dataset is selected for the experiment.

The dataset contains information such as title, content type, country, release year, rating, duration, genres, and date added.

```
from google.colab import files
import io
import os

uploaded = files.upload()

print("Uploaded files:")
for filename in uploaded.keys():
    print(filename)
```

```
Choose Files No file chosen  Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving netflix_titles.csv to netflix_titles.csv
Uploaded files:
netflix_titles.csv
```

```
# Find the uploaded CSV file
csv_files = [filename for filename in uploaded.keys() if filename.lower().endswith(".csv")]

if not csv_files:
    raise FileNotFoundError("No CSV file was uploaded.")

file_name = csv_files[0]

# Load dataset
df = pd.read_csv(io.BytesIO(uploaded[file_name]))

print("Dataset loaded successfully.")
print("File:", file_name)
print("Shape:", df.shape)

df.head()
```

```
Dataset loaded successfully.
File: netflix_titles.csv
Shape: (8807, 12)
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas,	NaN	September 24, 2021	2021	TV-MA	1 Season	Crime TV Shows, International	To protect his family from a powerful

Step 2: Analytical Objective

Business Objective

The objective of this analysis is to understand the composition and growth of Netflix's content library.

The analysis focuses on:

- The distribution of Movies and TV Shows.
- Content growth over time.
- Popular content ratings.
- Major content-producing countries.
- Content duration patterns.
- Relationships between important numerical variables.

The findings will be used to generate actionable business insights that can support content planning and strategic decision-making.

```
print("Number of rows:", df.shape[0])
print("Number of columns:", df.shape[1])
```

```
Number of rows: 8807
Number of columns: 12
```

```
print("Column names:")
print(df.columns.tolist())
```

```
Column names:
['show_id', 'type', 'title', 'director', 'cast', 'country', 'date_added', 'release_year', 'rating', 'duration', 'listed_in', 'de
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype 
---  ------        --------------  ----- 
 0   show_id       8807 non-null   object
 1   type          8807 non-null   object
 2   title         8807 non-null   object
 3   director      6173 non-null   object
 4   cast          7982 non-null   object
 5   country       7976 non-null   object
 6   date_added    8797 non-null   object
 7   release_year  8807 non-null   int64 
 8   rating        8803 non-null   object
 9   duration      8804 non-null   object
 10  listed_in     8807 non-null   object
 11  description   8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

```
df.describe(include="all").T
```

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
show_id	8807	8807	s8807	1	NaN	NaN	NaN	NaN	NaN	NaN	NaN
type	8807	2	Movie	6131	NaN	NaN	NaN	NaN	NaN	NaN	NaN
title	8807	8807	Zubaan	1	NaN	NaN	NaN	NaN	NaN	NaN	NaN
director	6173	4528	Rajiv Chilaka	19	NaN	NaN	NaN	NaN	NaN	NaN	NaN
cast	7982	7692	David Attenborough	19	NaN	NaN	NaN	NaN	NaN	NaN	NaN
country	7976	748	United States	2818	NaN	NaN	NaN	NaN	NaN	NaN	NaN
date_added	8797	1767	January 1, 2020	109	NaN	NaN	NaN	NaN	NaN	NaN	NaN
release_year	8807.0	NaN	NaN	NaN	2014.180198	8.819312	1925.0	2013.0	2017.0	2019.0	2021.0
rating	8803	17	TV-MA	3207	NaN	NaN	NaN	NaN	NaN	NaN	NaN
duration	8804	220	1 Season	1793	NaN	NaN	NaN	NaN	NaN	NaN	NaN
listed_in	8807	514	Dramas, International Movies	362	NaN	NaN	NaN	NaN	NaN	NaN	NaN
description	8807	8775	Paranormal activity at a lush, abandoned prope...	4	NaN	NaN	NaN	NaN	NaN	NaN	NaN

```
missing_summary = pd.DataFrame({
    "Missing Values": df.isnull().sum(),
    "Missing Percentage": (df.isnull().mean() * 100).round(2)
})

missing_summary.sort_values(
```

```
    by="Missing Values",
    ascending=False
)
```

	Missing Values	Missing Percentage
director	2634	29.91
country	831	9.44
cast	825	9.37
date_added	10	0.11
rating	4	0.05
duration	3	0.03
show_id	0	0.00
type	0	0.00
title	0	0.00
release_year	0	0.00
listed_in	0	0.00
description	0	0.00

```
print("Number of duplicate rows:", df.duplicated().sum())
```

```
Number of duplicate rows: 0
```

```
df.nunique().sort_values(ascending=False)
```

	0
show_id	8807
title	8807
description	8775
cast	7692
director	4528
date_added	1767
country	748
listed_in	514
duration	220
release_year	74
rating	17
type	2

dtype: int64

```
# Make a copy for analysis
eda_df = df.copy()

# Convert release year to numeric
eda_df["release_year"] = pd.to_numeric(
    eda_df["release_year"],
    errors="coerce"
)

# Convert date_added to datetime
eda_df["date_added"] = pd.to_datetime(
    eda_df["date_added"],
    errors="coerce"
)

# Extract year content was added to Netflix
eda_df["date_added_year"] = eda_df["date_added"].dt.year
```

```
# Extract first country
eda_df["primary_country"] = (
    eda_df["country"]
    .fillna("Unknown")
    .str.split(",")
    .str[0]
    .str.strip()
)

# Extract first genre/category
eda_df["primary_genre"] = (
    eda_df["listed_in"]
    .fillna("Unknown")
    .str.split(",")
    .str[0]
    .str.strip()
)

# Number of listed genres
eda_df["num_genres"] = (
    eda_df["listed_in"]
    .fillna("")
    .apply(lambda x: len([g for g in x.split(",") if g.strip()]))
)

# Number of countries associated with each title
eda_df["num_countries"] = (
    eda_df["country"]
    .fillna("")
    .apply(lambda x: len([c for c in x.split(",") if c.strip()]))
)

# Extract movie duration in minutes
eda_df["duration_minutes"] = np.where(
    eda_df["type"].eq("Movie"),
    pd.to_numeric(
        eda_df["duration"].str.extract(r"(\d+)")[0],
        errors="coerce"
    ),
    np.nan
)

# Extract TV show duration in seasons
eda_df["duration_seasons"] = np.where(
    eda_df["type"].eq("TV Show"),
    pd.to_numeric(
        eda_df["duration"].str.extract(r"(\d+)")[0],
        errors="coerce"
    ),
    np.nan
)

eda_df.head()
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	2021-09-25	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	2021-09-24	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	2021-09-24	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train I...

```
type_counts = eda_df["type"].value_counts()

print(type_counts)
```

```
type
Movie      6131
TV Show    2676
Name: count, dtype: int64
```

```
rating_counts = eda_df["rating"].value_counts()

print(rating_counts.head(10))
```

```
rating
TV-MA    3207
TV-14    2160
TV-PG     863
R         799
PG-13     490
TV-Y7     334
TV-Y      307
PG        287
TV-G      220
NR         80
Name: count, dtype: int64
```

```
country_counts = eda_df["primary_country"].value_counts()

print(country_counts.head(10))
```

```
primary_country
United States     3211
India             1008
Unknown            831
United Kingdom     628
Canada             271
Japan              259
France             212
South Korea        211
Spain              181
Mexico             134
Name: count, dtype: int64
```

```
genre_counts = eda_df["primary_genre"].value_counts()

print(genre_counts.head(10))
```

```
primary_genre
Dramas                      1600
Comedies                    1210
Action & Adventure           859
Documentaries                829
International TV Shows       774
Children & Family Movies     605
Crime TV Shows               399
Kids' TV                     388
Stand-Up Comedy              334
Horror Movies                275
Name: count, dtype: int64
```

```
total_titles = len(eda_df)

total_movies = (eda_df["type"] == "Movie").sum()

total_tv_shows = (eda_df["type"] == "TV Show").sum()

avg_release_year = eda_df["release_year"].mean()

print("KEY PERFORMANCE INDICATORS")
print("-" * 35)
print(f"Total Titles      : {total_titles:,}")
print(f"Total Movies      : {total_movies:,}")
print(f"Total TV Shows    : {total_tv_shows:,}")
print(f"Average Release Year: {avg_release_year:.0f}")
```

```
KEY PERFORMANCE INDICATORS
-----------------------------------
Total Titles      : 8,807
Total Movies      : 6,131
Total TV Shows    : 2,676
Average Release Year: 2014
```

```
kpi_html = f"""
<div style="display:flex; gap:15px; flex-wrap:wrap;">

<div style="padding:20px; border:1px solid #ccc; border-radius:10px; min-width:180px; text-align:center;">
<h3>Total Titles</h3>
<h2>{total_titles:,}</h2>
</div>

<div style="padding:20px; border:1px solid #ccc; border-radius:10px; min-width:180px; text-align:center;">
<h3>Movies</h3>
<h2>{total_movies:,}</h2>
</div>

<div style="padding:20px; border:1px solid #ccc; border-radius:10px; min-width:180px; text-align:center;">
<h3>TV Shows</h3>
<h2>{total_tv_shows:,}</h2>
</div>

<div style="padding:20px; border:1px solid #ccc; border-radius:10px; min-width:180px; text-align:center;">
<h3>Avg. Release Year</h3>
<h2>{avg_release_year:.0f}</h2>
</div>

</div>
"""

display(HTML(kpi_html))
```

Total Titles 8,807

Movies 6,131

TV Shows 2,676

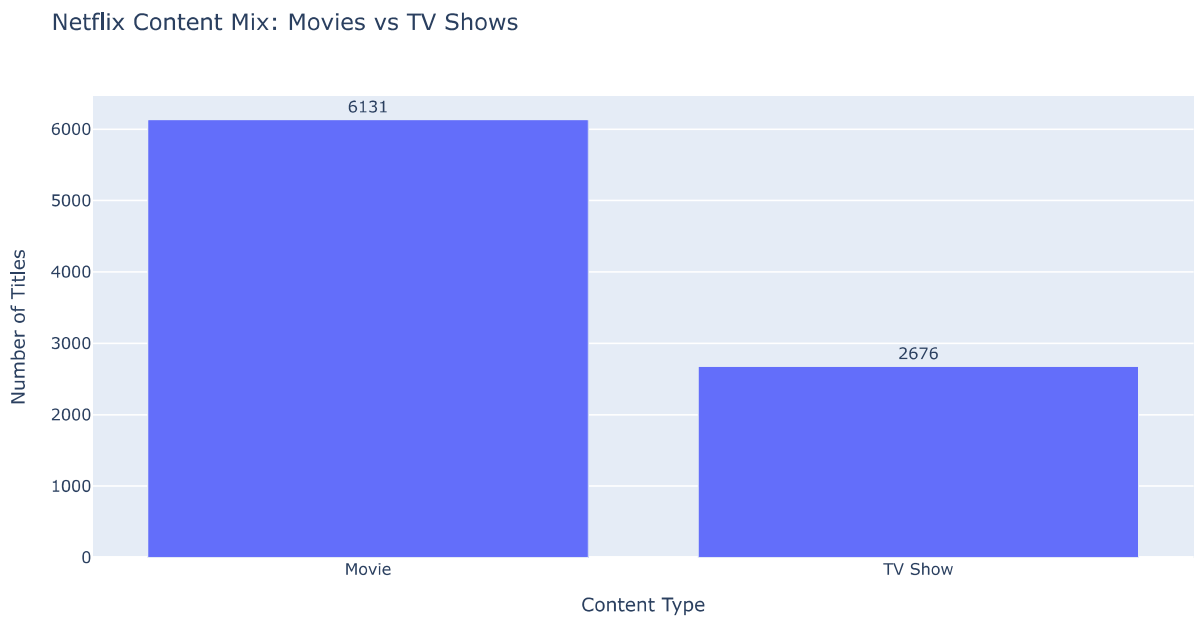
Avg. Release Year 2014

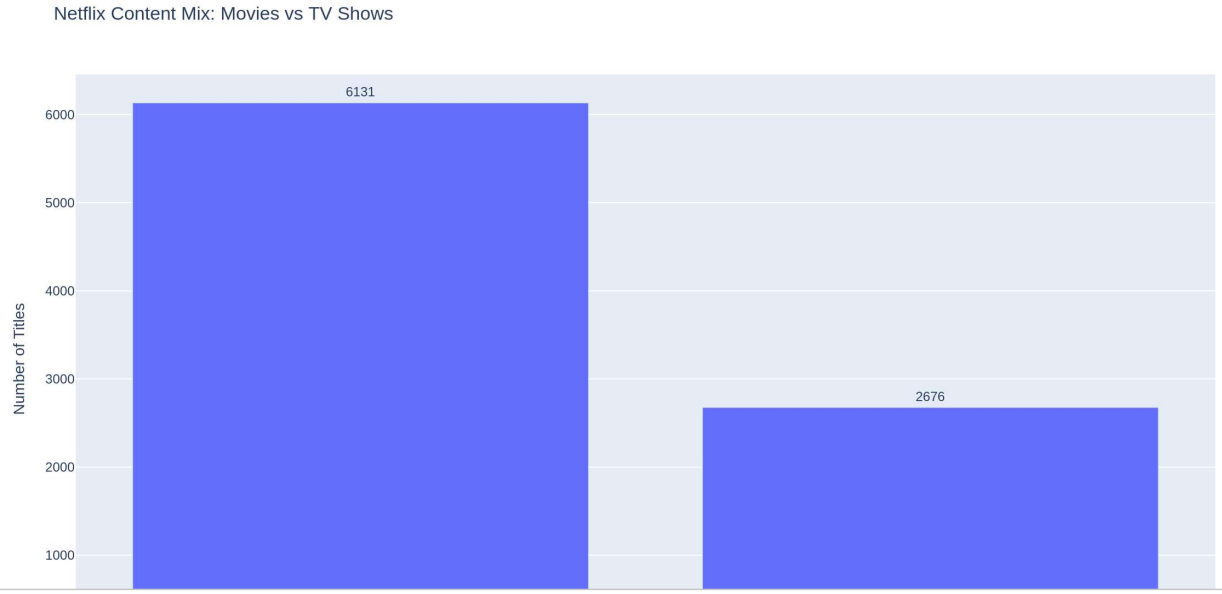
```
fig = px.bar(
    type_counts.reset_index(),
```

```
    x="type",
    y="count",
    title="Netflix Content Mix: Movies vs TV Shows",
    labels={
        "type": "Content Type",
        "count": "Number of Titles"
    },
    text="count"
)

fig.update_traces(textposition="outside")

fig.show()
display(Image(fig.to_image(format="png", width=1200, height=700, scale=2)))
```



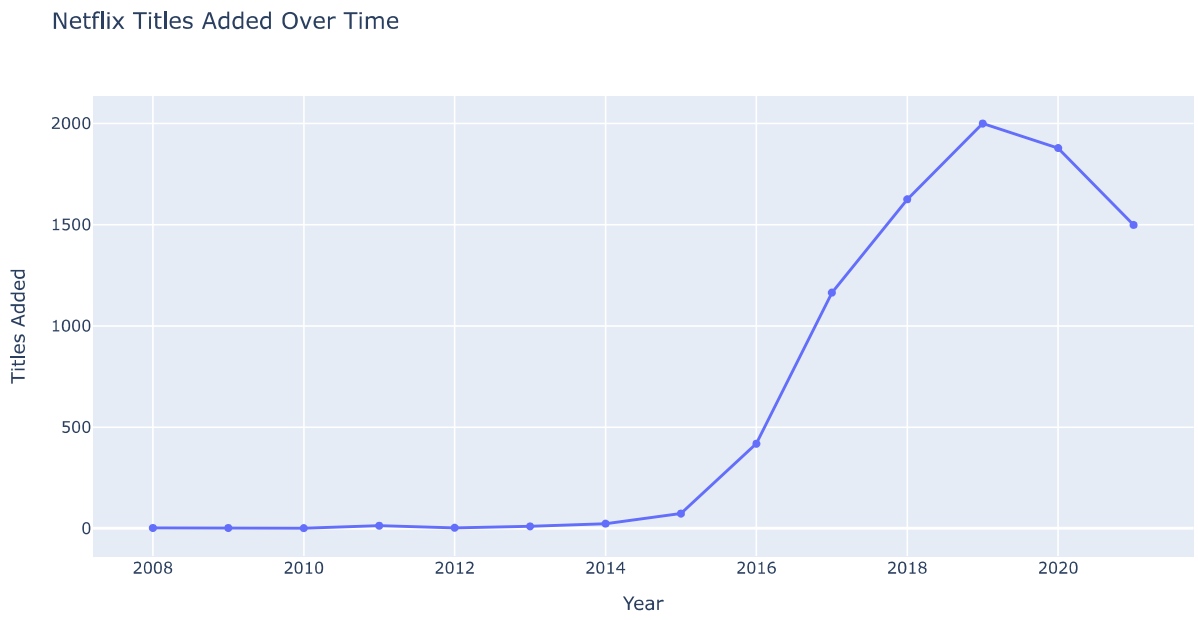


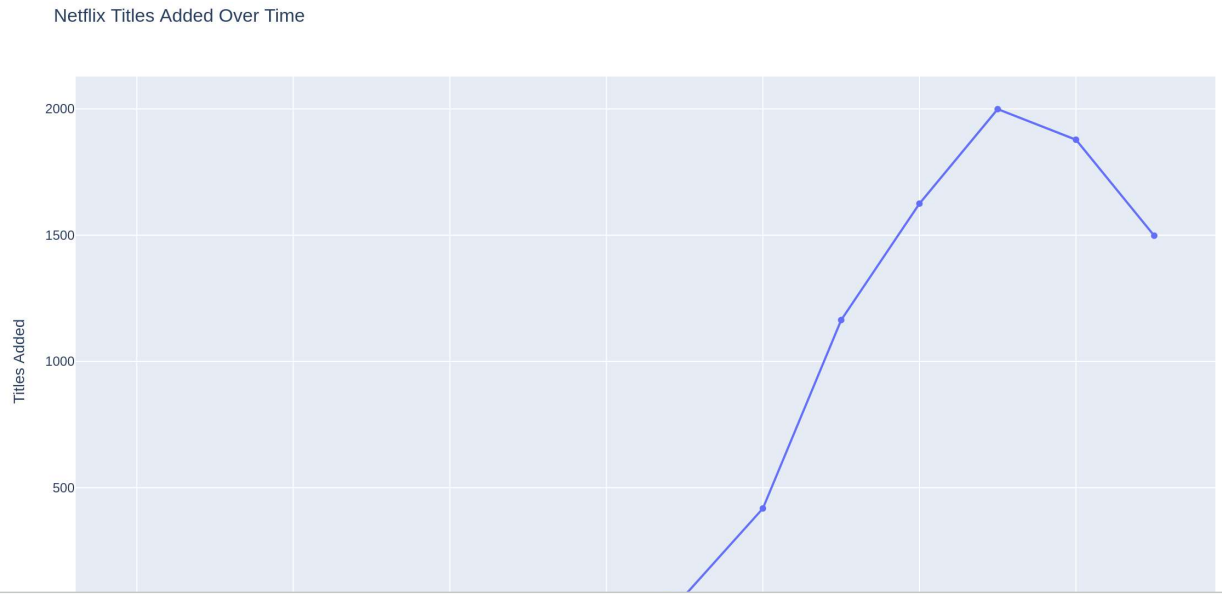

```
yearly_additions = (
    eda_df["date_added_year"]
    .dropna()
    .astype(int)
    .value_counts()
    .sort_index()
    .reset_index()
)
```

```
yearly_additions.columns = ["year", "titles_added"]

fig = px.line(
    yearly_additions,
    x="year",
    y="titles_added",
    markers=True,
    title="Netflix Titles Added Over Time",
    labels={
        "year": "Year",
        "titles_added": "Titles Added"
    }
)

fig.show()
display(Image(fig.to_image(format="png", width=1200, height=700, scale=2)))
```





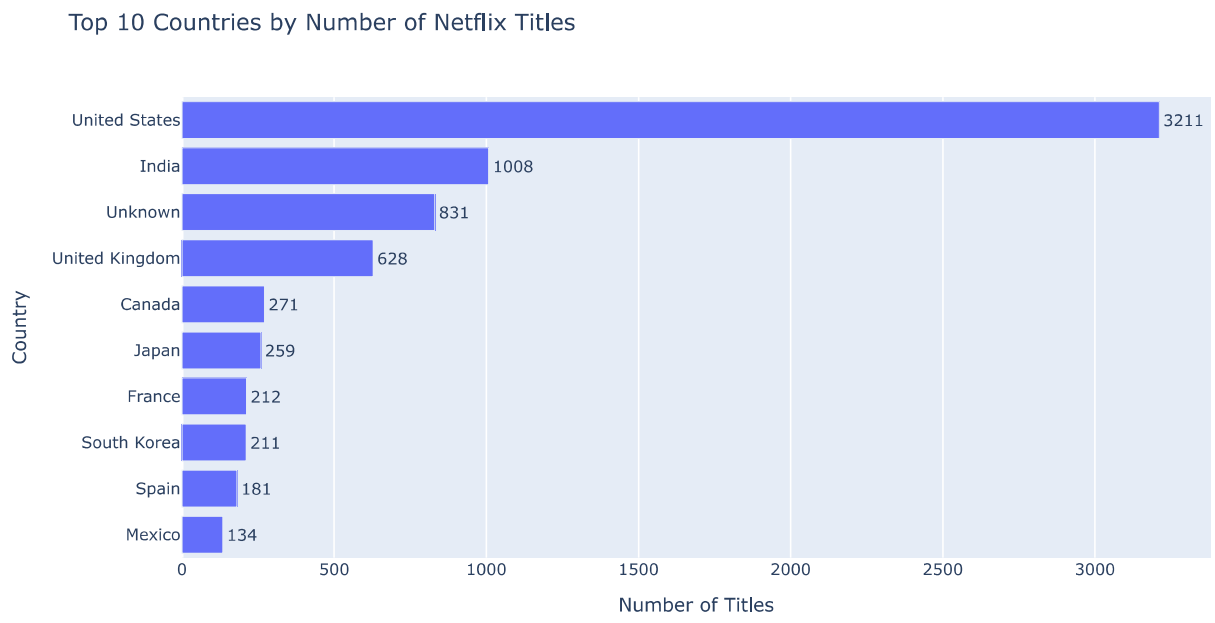

```
top_countries = (
    eda_df["primary_country"]
    .value_counts()
    .head(10)
    .sort_values()
    .reset_index()
)
```

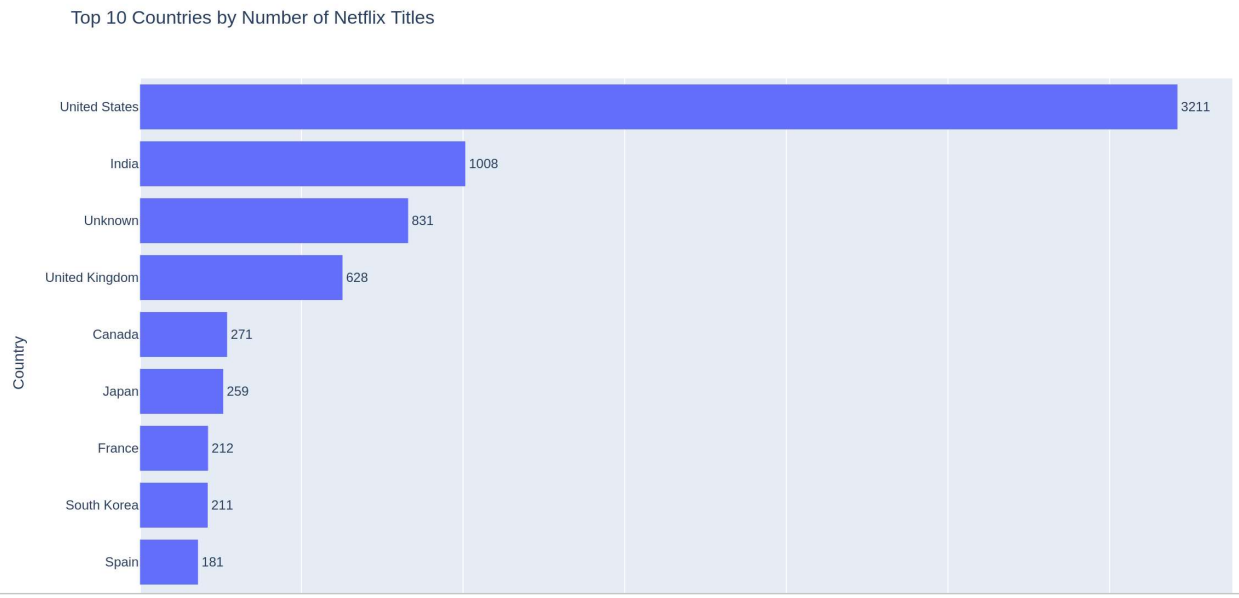
```
top_countries.columns = ["country", "titles"]

fig = px.bar(
    top_countries,
    x="titles",
    y="country",
    orientation="h",
    title="Top 10 Countries by Number of Netflix Titles",
    labels={
        "titles": "Number of Titles",
        "country": "Country"
    },
    text="titles"
)

fig.update_traces(textposition="outside")

fig.show()
display(Image(fig.to_image(format="png", width=1200, height=700, scale=2)))
```






```
top_ratings = (
    eda_df["rating"]
    .value_counts()
```

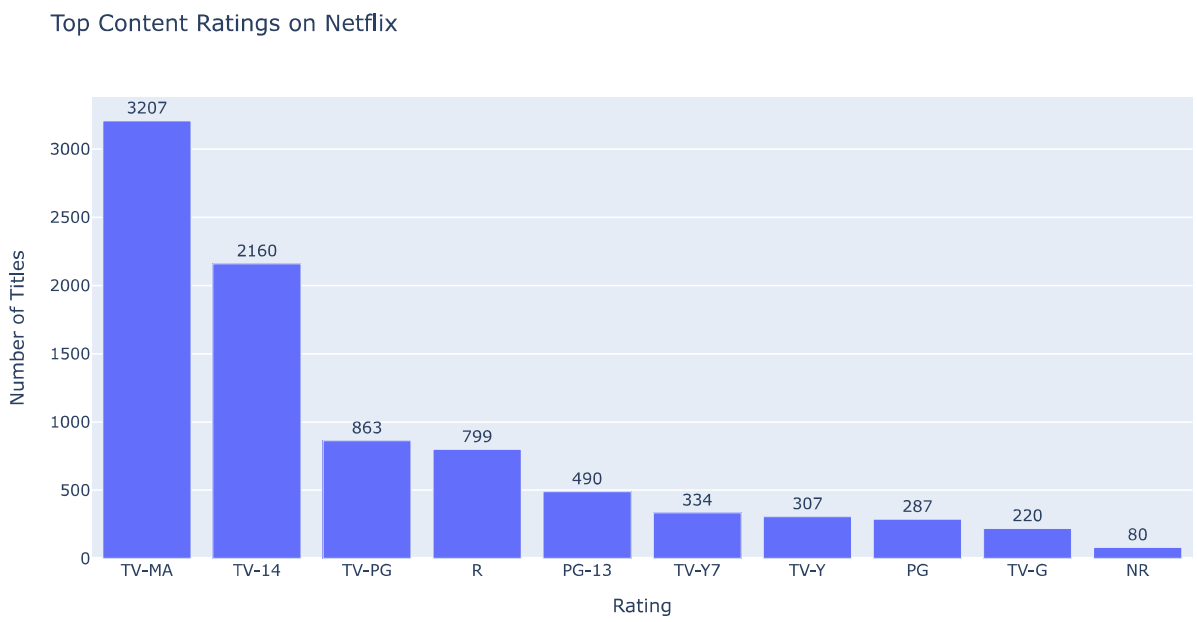
```
    .head(10)
    .reset_index()
)

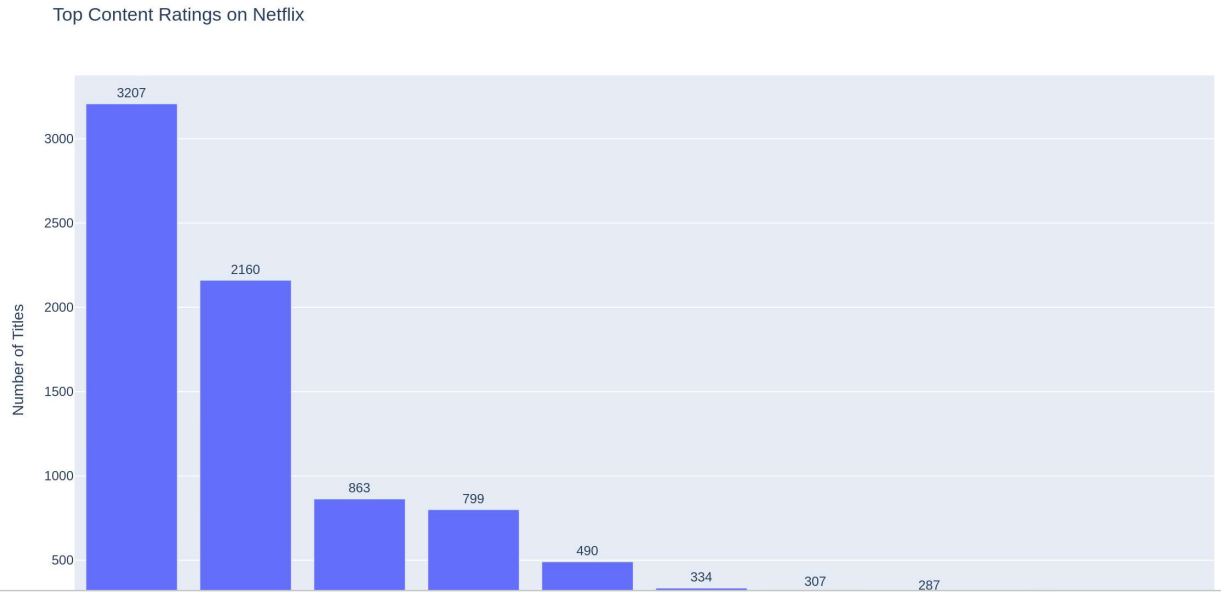
top_ratings.columns = ["rating", "titles"]

fig = px.bar(
    top_ratings,
    x="rating",
    y="titles",
    title="Top Content Ratings on Netflix",
    labels={
        "rating": "Rating",
        "titles": "Number of Titles"
    },
    text="titles"
)

fig.update_traces(textposition="outside")

fig.show()
display(Image(fig.to_image(format="png", width=1200, height=700, scale=2)))
```





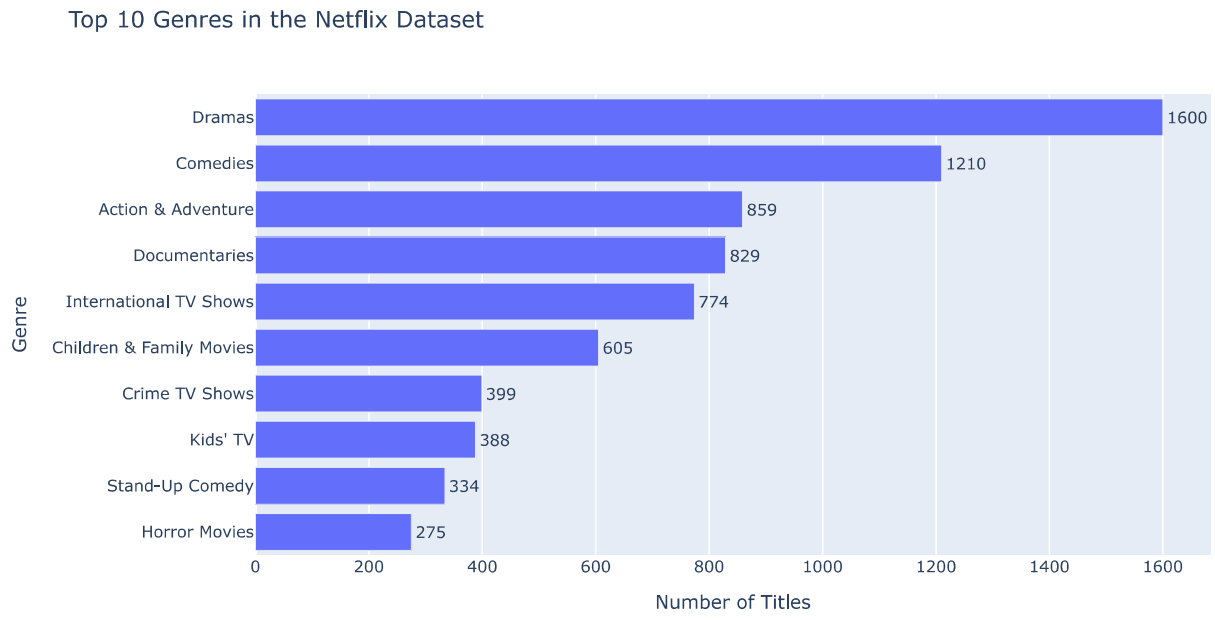
```
top_genres = (
    eda_df["primary_genre"]
    .value_counts()
    .head(10)
    .sort_values()
    .reset_index()
)

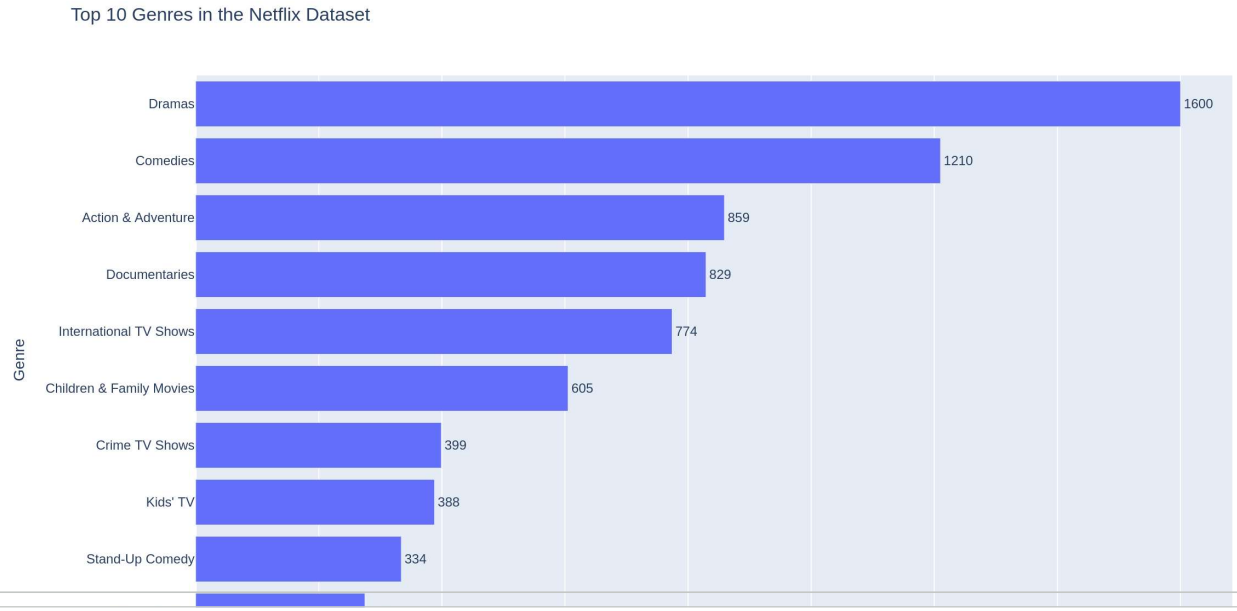
top_genres.columns = ["genre", "titles"]

fig = px.bar(
    top_genres,
    x="titles",
    y="genre",
    orientation="h",
    title="Top 10 Genres in the Netflix Dataset",
    labels={
        "titles": "Number of Titles",
        "genre": "Genre"
    },
    text="titles"
)

fig.update_traces(textposition="outside")

fig.show()
display(Image(fig.to_image(format="png", width=1200, height=700, scale=2)))
```

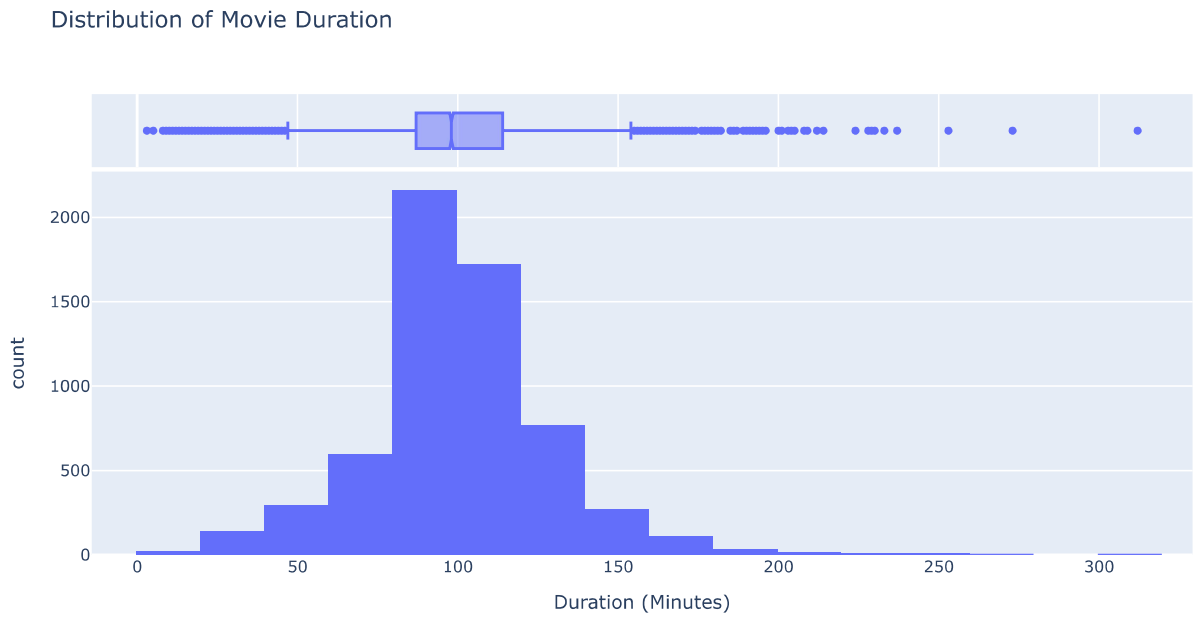





```
movie_duration = eda_df[
    eda_df["type"] == "Movie"
].dropna(subset=["duration_minutes"])

fig = px.histogram(
    movie_duration,
    x="duration_minutes",
    nbins=30,
    marginal="box",
    title="Distribution of Movie Duration",
    labels={
        "duration_minutes": "Duration (Minutes)"
    }
)

fig.show()
display(Image(fig.to_image(format="png", width=1200, height=700, scale=2)))
```

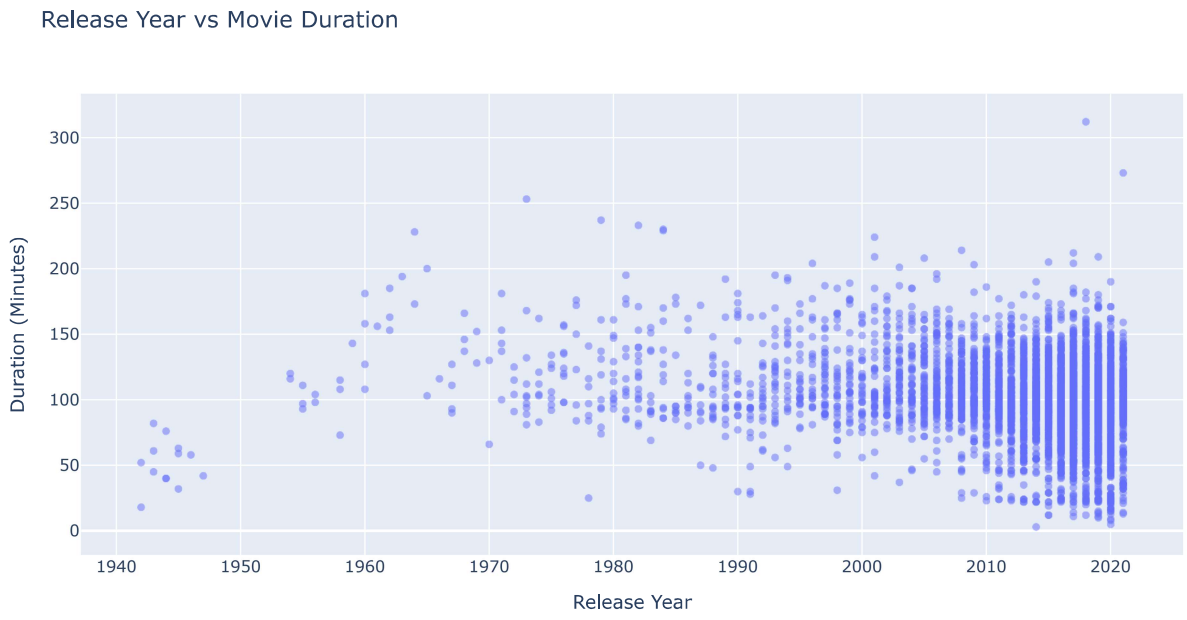


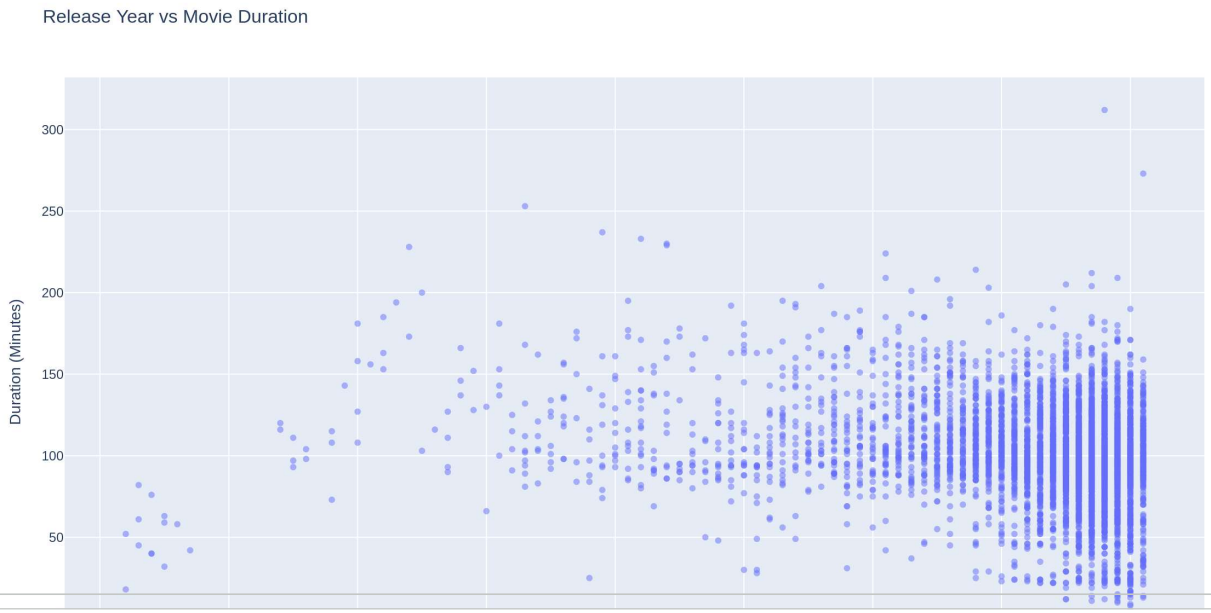



```
scatter_df = eda_df[
    (eda_df["type"] == "Movie") &
    eda_df["release_year"].notna() &
    eda_df["duration_minutes"].notna()
].copy()

fig = px.scatter(
    scatter_df,
    x="release_year",
    y="duration_minutes",
    hover_name="title",
    opacity=0.5,
    title="Release Year vs Movie Duration",
    labels={
        "release_year": "Release Year",
        "duration_minutes": "Duration (Minutes)"
    }
)

fig.show()
display(Image(fig.to_image(format="png", width=1200, height=700, scale=2)))
```






```
correlation_data = eda_df[
    [
        "release_year",
        "date_added_year",
        "duration_minutes",
        "duration_seasons",
        "num_genres",
        "num_countries"
    ]
]

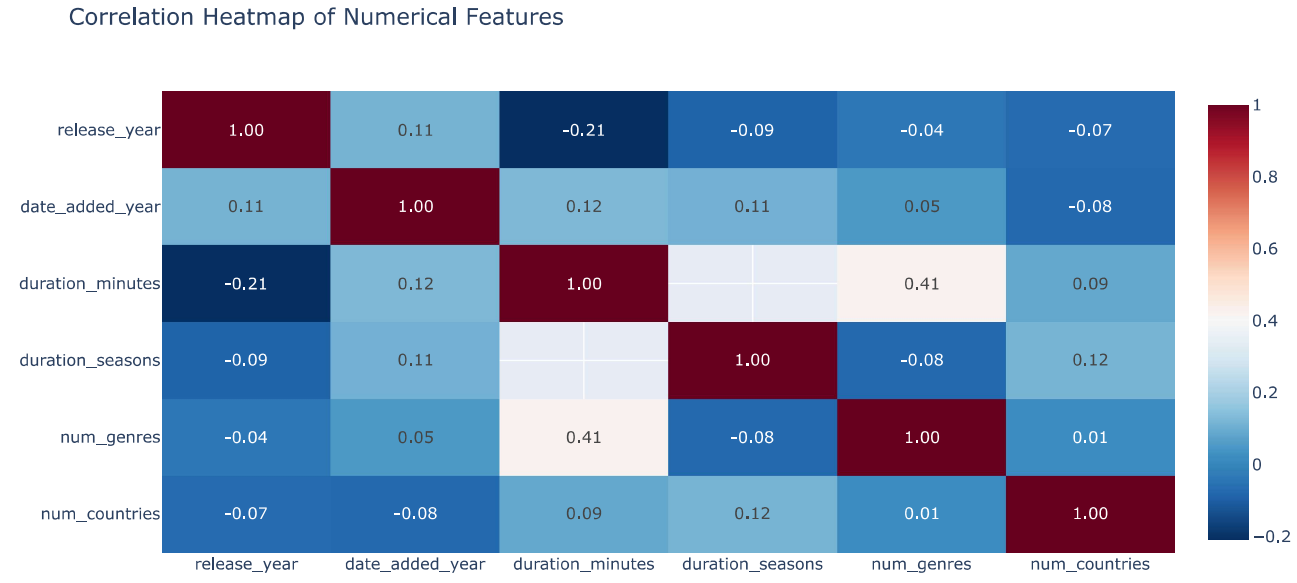
correlation_matrix = correlation_data.corr()

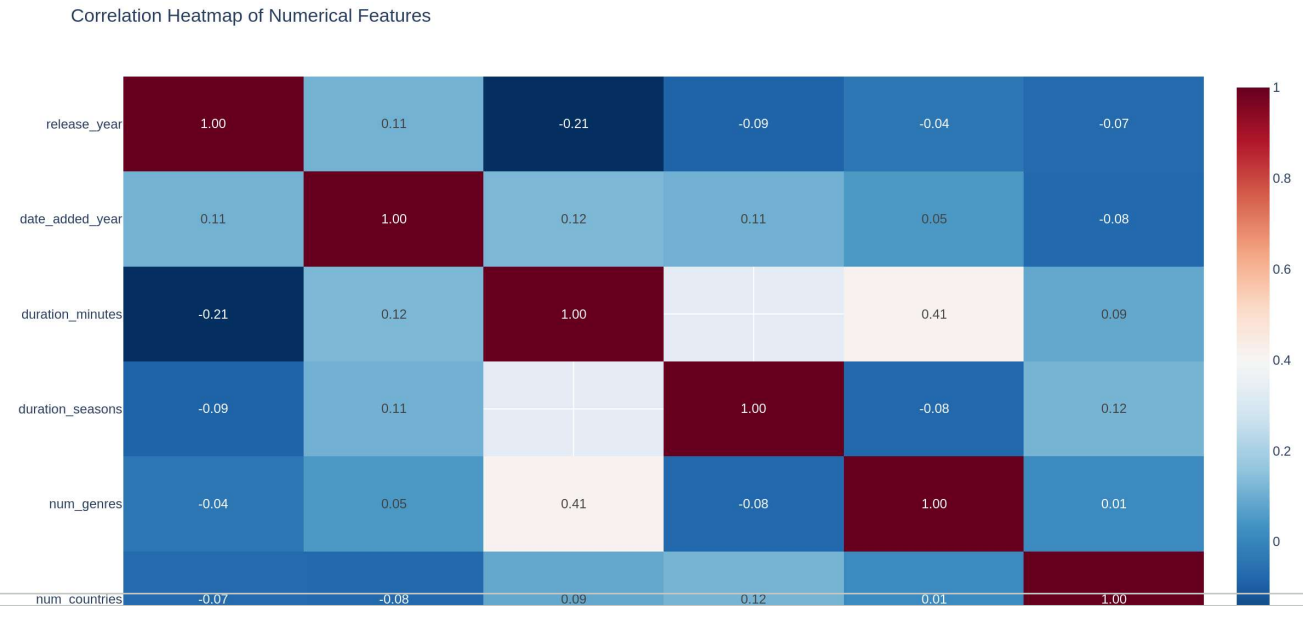
correlation_matrix
```

	release_year	date_added_year	duration_minutes	duration_seasons	num_genres	num_countries
release_year	1.000000	0.110473	-0.206285	-0.090194	-0.040877	-0.072946
date_added_year	0.110473	1.000000	0.124436	0.106543	0.048913	-0.075074
duration_minutes	-0.206285	0.124436	1.000000	NaN	0.413998	0.085912
duration_seasons	-0.090194	0.106543	NaN	1.000000	-0.077864	0.116376
num_genres	-0.040877	0.048913	0.413998	-0.077864	1.000000	0.011488
num_countries	-0.072946	-0.075074	0.085912	0.116376	0.011488	1.000000

```
fig = px.imshow(
    correlation_matrix,
    text_auto=".2f",
    aspect="auto",
    title="Correlation Heatmap of Numerical Features",
    color_continuous_scale="RdBu_r"
)

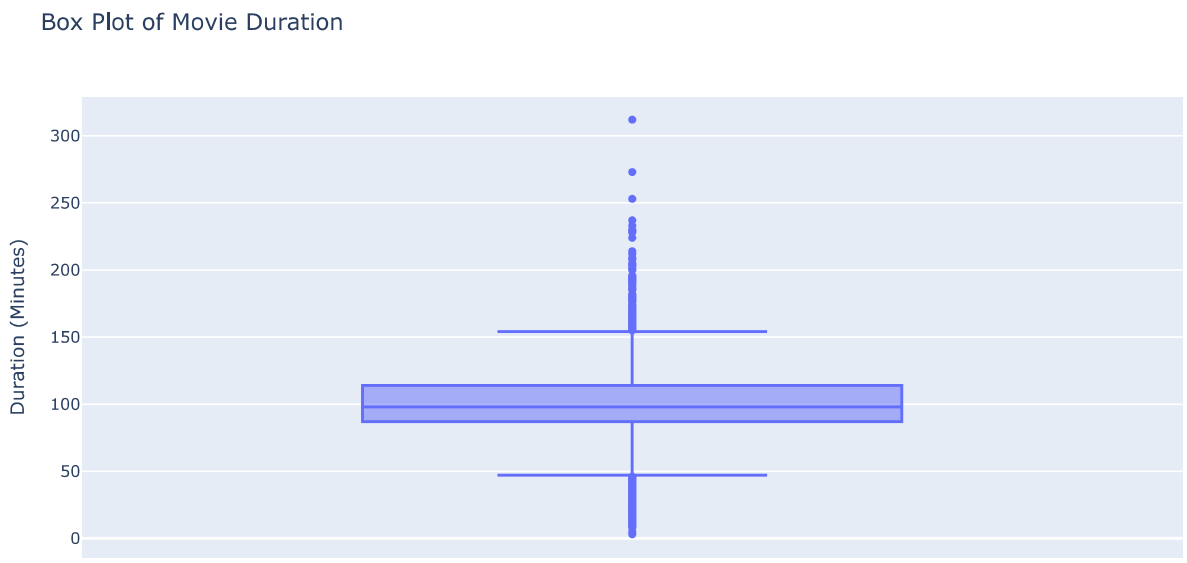
fig.show()
display(Image(fig.to_image(format="png", width=1200, height=700, scale=2)))
```





```
fig = px.box(
    movie_duration,
    y="duration_minutes",
    title="Box Plot of Movie Duration",
    labels={
        "duration_minutes": "Duration (Minutes)"
    }
)

fig.show()
display(Image(fig.to_image(format="png", width=1200, height=700, scale=2)))
```





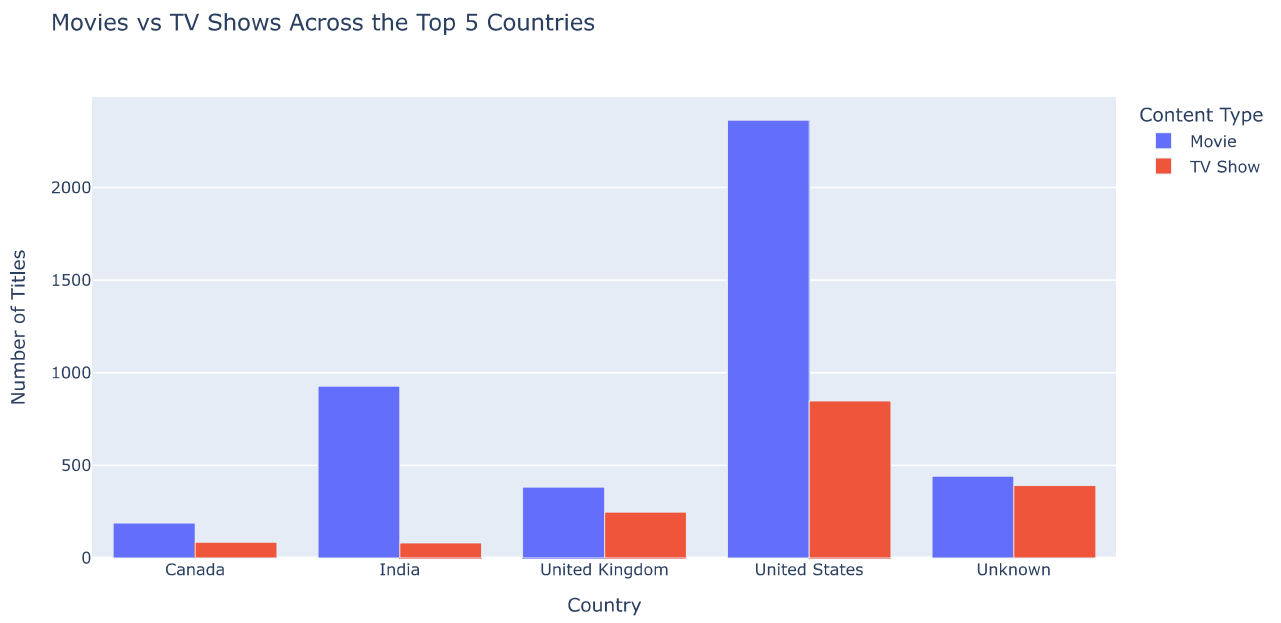

```
top_5_countries = (
    eda_df["primary_country"]
    .value_counts()
    .head(5)
    .index
)

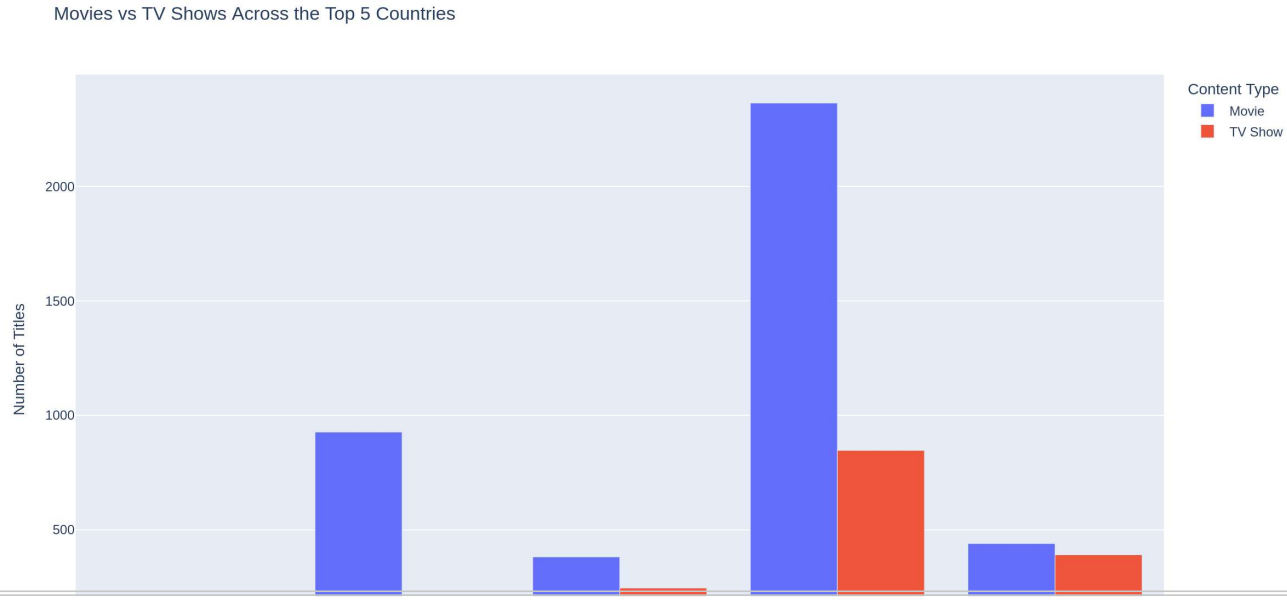
country_type_df = eda_df[
    eda_df["primary_country"].isin(top_5_countries)
]
```

```
country_type_counts = (
    country_type_df
    .groupby(["primary_country", "type"])
    .size()
    .reset_index(name="titles")
)

fig = px.bar(
    country_type_counts,
    x="primary_country",
    y="titles",
    color="type",
    barmode="group",
    title="Movies vs TV Shows Across the Top 5 Countries",
    labels={
        "primary_country": "Country",
        "titles": "Number of Titles",
        "type": "Content Type"
    }
)

fig.show()
display(Image(fig.to_image(format="png", width=1200, height=700, scale=2)))
```





```
# Prepare data for dashboard

# Content type
type_dashboard = (
    eda_df["type"]
    .value_counts()
    .reset_index()
)
type_dashboard.columns = ["type", "titles"]

# Yearly additions
year_dashboard = (
    eda_df["date_added_year"]
    .dropna()
    .astype(int)
    .value_counts()
    .sort_index()
    .reset_index()
)
year_dashboard.columns = ["year", "titles"]

# Top countries
country_dashboard = (
    eda_df["primary_country"]
    .value_counts()
    .head(8)
    .sort_values()
    .reset_index()
)
country_dashboard.columns = ["country", "titles"]

# Ratings
rating_dashboard = (
    eda_df["rating"]
    .value_counts()
    .head(8)
    .sort_values()
    .reset_index()
)
rating_dashboard.columns = ["rating", "titles"]


# Create dashboard
fig = make_subplots(
    rows=2,
    cols=2,
    subplot_titles=(
        "Movies vs TV Shows",
        "Titles Added Over Time",
        "Top Countries",
        "Top Ratings"
    )
)

# Chart 1
fig.add_trace(
    go.Bar(
        x=type_dashboard["type"],
        y=type_dashboard["titles"],
        name="Content Type"
    ),
    row=1,
    col=1
)

# Chart 2
fig.add_trace(
    go.Scatter(
        x=year_dashboard["year"],
        y=year_dashboard["titles"],
        mode="lines+markers",
        name="Titles Added"
    ),
    row=1,
    col=2
)
```

```
# Chart 3
fig.add_trace(
    go.Bar(
        x=country_dashboard["titles"],
        y=country_dashboard["country"],
        orientation="h",
        name="Countries"
    ),
    row=2,
    col=1
)

# Chart 4
fig.add_trace(
    go.Bar(
        x=rating_dashboard["rating"],
        y=rating_dashboard["titles"],
        name="Ratings"
    ),
    row=2,
    col=2
)

fig.update_layout(
    title_text="Netflix Data Story Dashboard",
    height=800,
    showlegend=False
)

fig.update_xaxes(title_text="Content Type", row=1, col=1)
fig.update_yaxes(title_text="Titles", row=1, col=1)

fig.update_xaxes(title_text="Year", row=1, col=2)
fig.update_yaxes(title_text="Titles Added", row=1, col=2)

fig.update_xaxes(title_text="Titles", row=2, col=1)
fig.update_yaxes(title_text="Country", row=2, col=1)

fig.update_xaxes(title_text="Rating", row=2, col=2)
fig.update_yaxes(title_text="Titles", row=2, col=2)

fig.show()
display(Image(fig.to_image(format="png", width=1200, height=700, scale=2)))
```




```
# Most common content type
most_common_type = eda_df["type"].value_counts().idxmax()

# Most common country
most_common_country = eda_df["primary_country"].value_counts().idxmax()

# Most common rating
most_common_rating = eda_df["rating"].value_counts().idxmax()

# Most common genre
most_common_genre = eda_df["primary_genre"].value_counts().idxmax()

# Year with maximum additions
year_with_max_additions = (
    eda_df["date_added_year"]
    .dropna()
    .astype(int)
    .value_counts()
    .idxmax()
)

# Average movie duration
average_movie_duration = eda_df["duration_minutes"].mean()
```

```
print("KEY FINDINGS")
print("=" * 50)
print(f"Most common content type : {most_common_type}")
print(f"Top primary country      : {most_common_country}")
print(f"Most common rating       : {most_common_rating}")
print(f"Most common genre        : {most_common_genre}")
print(f"Peak year of additions   : {year_with_max_additions}")
print(f"Average movie duration   : {average_movie_duration:.1f} minutes")
```

```
KEY FINDINGS
==================================================
Most common content type : Movie
Top primary country      : United States
Most common rating       : TV-MA
Most common genre        : Dramas
Peak year of additions   : 2019
Average movie duration   : 99.6 minutes
```

Data Story

1. Business Context

Netflix operates a large content library containing both Movies and TV Shows. Understanding the structure and evolution of this library is important for content planning and strategic decision-making.

2. Key Metrics

The analysis begins with total titles, Movies, TV Shows, and the average release year to establish the overall scale and composition of the dataset.

3. Content Trend

The yearly trend visualization shows how the Netflix content library has changed over time and highlights periods of higher content additions.

4. Geographic Distribution

The country analysis shows which countries contribute the largest number of titles to the dataset.

5. Audience and Content Profile

The rating and genre visualizations help identify the dominant categories of content in the Netflix library.

6. Content Characteristics

Movie duration analysis shows the typical duration of movies and helps identify unusually short or long titles.

7. Relationships and Patterns

The scatter plot and correlation heatmap help examine relationships between numerical variables and identify patterns in the dataset.

8. Business Implications

The findings can help decision-makers understand content composition, geographic concentration, audience categories, and changes in the content library over time.

Actionable Recommendations

1. Monitor content additions over time to identify periods of strong or weak content growth and support future content planning.
2. Use country-level content patterns to understand geographic concentration and identify opportunities for broader regional representation.
3. Analyze dominant content ratings and genres when planning future acquisitions or productions.
4. Use movie-duration patterns to understand audience content preferences and maintain an appropriate mix of shorter and longer content.

5. Regularly update the analysis as new content is added so that business decisions remain aligned with current content trends.

Final Business Dashboard

The dashboard combines the major KPIs and visualizations into a single interactive view.

The dashboard helps stakeholders quickly understand:

- Overall content volume
- Movies versus TV Shows
- Content growth over time
- Geographic distribution
- Content-rating patterns
- Major trends and business opportunities